# ShahLab.AI Phase 2 Proposal — CLARA

Clinical Learning, Assessment & Review Assistant (local reference PDF)

ShahLab.AI
CLARA | CLINICAL LEARNING, ASSESSMENT & REVIEW ASSISTANT
AI-Powered Clinical Skill Validation to Accelerate the Nursing Workforce
Shen Chang • Dr. Nan Kong | Purdue University, Biomedical Engineering | Clinical partners: Purdue School of Nursing■NCNC
Societal Challenge
Partner Organization
AI Technology
Expected Impact
Clinical skills assessment struggles to keep pace with nursing training demand.
Purdue School of Nursing; North Central Nursing Clinics
CV + MLLMs + Ontology-RAG + TAS + Procedural Modeling
≥70% less faculty review time; ≥95% critical-error agreement (pilot targets).

THE CHALLENGE

The U.S. nursing shortage is not only a workforce problem but a training capacity crisis. In 2025, nursing schools turned away 93,176 qualified applications, up from 80,162 in 2024 [1,2], largely due to limited faculty and clinical training resources [1]. One time-intensive requirement is the clinical skills “checkoff,” which requires individual competency validation. For a cohort of 40 students, validating a single eight-minute procedure takes over five hours of faculty observation (40 × 8 min ≈ 5.3 hours), excluding setup, instructor debriefing, and retakes.

Time is only part of the problem. Clinical competence requires recognizing subtle errors, such as sterile-field breaches or incorrect landmarking, and evaluating them in context. These assessments demand expert judgment and individualized feedback, making it difficult for educators to maintain consistent evaluation quality as student enrollment grows. Without scalable assessment support, institutions face a difficult tradeoff between training capacity and clinical competency assurance. This raises a critical question: How can we scale skills validation while preserving feedback quality and patient safety?

OUR AI SOLUTION

Fig. 1. CLARA Three-Tier Annotation System

- Correctness - Whether each step is performed accurately according to clinical standards
- Performance - How effectively and smoothly each step is executed
- Allowance - External factors that may affect task execution

Fig. 2. CLARA Platform Interface Demo

WORKING WITH PARTNERS

| Purdue Nursing | Data & Lab Validation
Connection: Libby Richards, School Head; Dagne Hartman
Continue nursing video collection, expert annotation, and rubric development. Faculty and students will lead initial CLARA testing, sharing insights to refine clinical assessments.

| NCNC | Clinical Validation
Connection: Jennifer Coddington, Medical Director
Clinical preceptors will evaluate CLARA in practice-based training, validate assessment quality, and identify safety concerns and improvements for clinical workflows.

PHASE 2 GOALS

The requested $50k in ShahLab.AI Phase 2 funding will support platform engineering,

student-facing web development, community pilot testing, expert-led clinical validation, and deployment preparation over 12 months.

Months 1–3 | Platform Development:

- Convert the existing annotation tool and AI models into an automated video analysis pipeline.
- Develop a student-facing web portal for video uploads, step-level assessment reports, and actionable AI feedback.

Months 3–6 | Community Pilot Testing:

- Deploy a supervised pilot with a target of 20–30 nursing students and 10–12 instructors/preceptors at Purdue Nursing and NCNC.
- Gather user perspectives to evaluate platform usability, clinical relevance, and learning experience.

Months 6–9 | Clinical Validation & AI Refinement:

- Benchmark CLARA against expert assessments, measuring accuracy, safety-critical error detection, clinical guidance quality, and faculty time savings.
- Incorporate expert-validated corrections into the clinical RAG knowledge base through human-in-the-loop refinement.

Months 9–12 | Strategic Partnerships & Sustainability:

- Strengthen CLARA's deployment readiness through platform improvements, data privacy safeguards, instructor oversight protocols, and faculty onboarding resources.
- Leverage Purdue Nursing Lab's existing relationship with ATI to explore integration into its established educational ecosystem.
- Engage with Laerdal to assess SimCapture compatibility and explore pilot collaborations and licensing opportunities to support broader adoption beyond ShahLab.AI funding.

To address the need, we have developed CLARA (Clinical Learning and Assessment Reasoning Agent), an MLLM-powered nursing skills assessment system. Built on authentic student skill validation videos and expert-grounded annotations, CLARA is designed to learn from real-world errors and subtle “grey-area” deviations. By combining procedural video understanding with multidimensional assessment of correctness, execution quality, and contextual difficulty, CLARA moves beyond basic step recognition toward fine-grained clinical performance evaluation. These assessments are further grounded in instructor-defined checkoff rubrics and expert reviews to provide explainable evaluations and personalized, actionable feedback (Figs. 1–2).

In practice, CLARA supports the clinical learning process through three interconnected functions: assessing performance, guiding skill improvement, and tracking learning progress over time.

01 Examiner

02 Personal Coach

03 Learning Diary

Step & error assessment

Actionable, skill-specific feedback

Longitudinal skill progress tracking

Together, these functions create a continuous learning cycle, empowering nursing students to practice independently, learn from mistakes, and progressively strengthen their clinical skills.

PROGRESS SINCE PHASE 1

Phase 1

Phase 2 — Proposed

AI Development:

Developed a first-of-its-kind nursing video dataset capturing authentic procedural errors, an expert annotation platform, and trained procedural assessment AI models.

Clinical Validation:

Evaluate CLARA with nursing students, instructors, and NCNC practitioners, measuring assessment accuracy, feedback quality, and faculty workload.

Community Partnership:

Collaborated with Purdue School of Nursing to collect authentic skill-validation videos and

establish expert-grounded assessment standards.
Community-Driven Refinement:
Incorporate student and preceptor input through human-in-the-loop RAG updates, supporting continuous improvement and broader adoption.

## WHY IT MATTERS

### PEOPLE

Students gain specific coaching on incomplete or unsafe actions. Preceptors focus attention on uncertain or high-risk steps, while retaining all formal evaluation decisions. Longer term, structured refreshers could help nurses with limited supervision.

### EXPECTED IMPACTS

CLARA aims to reduce faculty review time by ≥70% while achieving ≥95% agreement on safety-critical error detection, enabling more efficient and consistent nursing skills assessment. Supported by preliminary results of 92.4% step-recognition accuracy and 85.1% agreement with instructor-annotated assessments, CLARA has the potential to expand training capacity and personalized feedback without increasing faculty workload.

### PATH TO SCALE

Building on Phase 1's research foundation and Phase 2's planned clinical validation, CLARA will progress from a research prototype toward institutional deployment and broader healthcare adoption. Its web-based architecture will enable nursing programs and healthcare institutions to integrate AI-assisted skills assessment into existing training workflows, expanding access to personalized clinical learning without proportionally increasing faculty workload.
For long-term sustainability, we envision two complementary revenue models: institutional subscriptions for nursing schools, clinics, and hospitals, and platform licensing through established educational technology providers. Potential collaborations with ATI and Laerdal could accelerate regional and national adoption through existing institutional networks.
Beyond nursing education, our vision is to build a connected learning community where healthcare professionals can practice, receive personalized feedback, and continuously improve their skills wherever they are. Ultimately, we envision a future where every healthcare professional can train with insight, grow with confidence, and deliver better care.

References:
[1] American Association of Colleges of Nursing (AACN), “Schools of Nursing See Enrollment Increases Across Most Program Levels, Signaling Strong Interest in Nursing Careers,” press release, May 7, 2026. Data from 2025-2026 Enrollment and Graduations in Baccalaureate and Graduate Programs in Nursing.
[2] AACN, “Schools of Nursing Enrollment Increases Across Most Program Levels, Signaling Strong Interest in Nursing Careers,” press release, June 17, 2025. Data from 2024-2025 Enrollment and Graduations in Baccalaureate and Graduate Programs in Nursing.